Problem Statement 1 :- GlobeTrotter - Empowering Personalized Travel Planning

Overall Vision

The overarching vision for GlobeTrotter is to become a personalized, intelligent, and collaborative platform that transforms the way individuals plan and experience travel. The platform aims to empower users to dream, design, and organize trips with ease by offering an end-to-end travel planning tool that combines flexibility and interactivity.

It envisions a world where users can explore global destinations, visualize their journeys through structured itineraries, make cost-effective decisions, and share their travel plans within a community—making travel planning as exciting as the trip itself.

Mission

The mission for the hackathon team is to build a user-centric, responsive application that simplifies the complexity of planning multi-city travel. This involves constructing a solution that is both functional and insightful, powered by a well-designed relational database and a smooth frontend experience. Teams must focus on enabling users to organize personalized trips efficiently, stay within budget, and enjoy full visibility of their journey.

This involves constructing a solution that is both functional and insightful, powered by a well-designed relational database and a smooth frontend experience. Teams must focus on enabling users to organize personalized trips efficiently, stay within budget, and enjoy full visibility of their journey.

Problem Statement

Design and develop a complete travel planning application where users can:

- Create customized multi-city itineraries
- Assign travel dates, activities, and budgets
- Discover activities and destinations through search
- Receive cost breakdowns and visual calendars
- Share their plans publicly or with friends

The application must demonstrate proper use of relational databases to store and retrieve complex travel data such as user-specific itineraries, stops, activities, and estimated expenses. The system should also support dynamic user interfaces that adapt to each user's trip flow.

The application for GlobalTrotters will include the following comprehensive set of features, ensuring a rich and user-friendly experience across desktop or mobile platforms:

1. Login / Signup Screen

Description: Entry point of the app allowing users to create or access their account.

Purpose: Authenticate users to manage personal travel plans.

Key Functionality/Components:

- Email & password fields
- Login button
- Signup link
- "Forgot Password"
- Basic validation

2. Dashboard / Home Screen

Description: Central hub showing upcoming trips, popular cities, and quick actions.

Purpose: Allows users to navigate to their trips and explore inspiration.

Key Functionality/Components:

- Welcome message
- List of recent trips
- “Plan New Trip” button
- Recommended destinations
- Budget highlights

3. Create Trip Screen

Description: Form to initiate a new trip by providing a name, travel dates, and a description.

Purpose: Begins the process of creating a personalized travel plan.

Key Functionality/Components:

- Trip name
- Start & end dates
- Trip description
- Cover photo upload (optional)
- Save button

4. My Trips (Trip List) Screen

Description: List view of all trips created by the user with basic summary data.

Purpose: Easily access and manage existing or upcoming trips.

Key Functionality/Components:

- Trip cards showing name
- Date range
- Destination count
- Edit/View/Delete actions

5. Itinerary Builder Screen

Description: Interface to add cities, dates, and activities for each stop.

Purpose: Construct the full day-wise trip plan in an interactive format.

Key Functionality/Components:

- “Add Stop” button
- Select city and travel dates
- Assign activities to each stop
- Reorder cities

6. Itinerary View Screen

Description: Visual representation of the completed trip itinerary.

Purpose: Review the full plan in a structured format.

Key Functionality/Components:

- Day-wise layout
- City headers
- Activity blocks with time and cost
- View mode toggle (calendar/list)

7. City Search

Description: Search interface to find and add cities to a trip, with information like country, cost index, and popularity.

Purpose: Discover and include relevant cities in the itinerary.

Key Functionality/Components:

- Search bar
- List of cities with meta info
- “Add to Trip” button
- Filter by country/region

8. Activity Search

Description: Browse and select things to do in each stop, categorized by interest or cost.

Purpose: Enrich trips with experiences like sightseeing, food tours, or adventure activities.

Key Functionality/Components:

- Activity filters (type, cost, duration)
- Add/remove buttons
- Quick view of description and images

9. Trip Budget & Cost Breakdown Screen

Description: Summarized financial view showing estimated total cost and breakdowns.

Purpose: Helps travelers stay informed and within budget.

Key Functionality/Components:

- Cost breakdown by transport, stay, activities, meals
- Pie/bar charts
- Average cost per day
- Alerts for overbudget days

10. Trip Calendar / Timeline Screen

Description: Calendar-based or vertical timeline view of the full itinerary.

Purpose: Helps users visualize the journey and daily plan flow.

Key Functionality/Components:

- Calendar component
- Expandable day views
- Drag-to-reorder activities
- Quick editing options

11. Shared/Public Itinerary View Screen

Description: Public page displaying a sharable version of an itinerary.

Purpose: Allows others to view, get inspired, or copy the trip.

Key Functionality/Components:

- Public URL

- Itinerary summary
- “Copy Trip” button
- Social media sharing
- Read-only view

12. User Profile / Settings Screen

Description: User settings page to update profile information and preferences.

Purpose: Enables users to control their data, preferences, and privacy.

Key Functionality/Components:

- Editable fields (name, photo, email)
- Language preference
- Delete account
- Saved destinations list

13. Admin / Analytics Dashboard (Optional)

Description: Admin-only interface to track user trends, trip data, and platform usage.

Purpose: Helps in monitoring app adoption, popular cities, and user behavior.

Key Functionality/Components:

- Tables and charts of trips created
- Top cities/activities
- User engagement stats
- User management tools

AI/GenAI Enhancement

The application should include an AI-powered travel planning component with the following additional functionalities:

1. **AI Travel Assistant:** Allow users to describe their travel requirements in natural language, such as destination, duration, budget, interests, and number of travelers. The system should generate a personalized itinerary based on the provided requirements.
2. **AI Itinerary Optimization:** Allow the AI to analyze an existing itinerary and suggest improvements based on factors such as budget, activity preferences, travel duration, and the sequence of destinations.
3. **Conversational Trip Assistant:** Provide a conversational interface where users can ask questions about their planned trip and receive context-aware recommendations for destinations, activities, and budget planning.

Teams may use suitable LLMs, APIs, RAG, AI agents, or other GenAI approaches to implement these functionalities.

Problem Statement 2 :- Rental Management

The Rental Management application streamlines the entire rental process. It provides a unified platform to manage products, schedule pickups. Online booking and reservations are done through a website portal. Customers can shop available products, reserve their selected dates, and checkout right on the website.

Whether you rent by the hour, week, month or year, create customized pricing for every product based on different time frames.

Roles

- Customer
- End User

1. Rental Product Management

- Define Rentable Products: Mark products as rentable and configure units (e.g., per hour, day, week).
- Custom Rental Duration: Supports short-term and long-term rentals with flexible durations.
- Product Availability: View product availability in calendar or list view to avoid overbooking.

2. Rental Quotations & Orders

- Create rental quotations, confirm them into rental orders, and generate rental contracts.
- Customers can review, confirm, and pay online through the customer portal.
- Pickup/Return Scheduling: Track and schedule pickup and return with precise timing.

Order and Rental Flow

As soon as you confirm a rental order, the system “reserves” the required items.

This means those products are set aside and cannot be given to anyone else.

It ensures you always have the right items ready for that customer.

3. Delivery Management

Reservation (Order Confirmed)

Pickup (Delivery to the Customer)

The system creates a pickup document telling your pickup team which items need to be delivered.

This makes it easy to prepare and hand over the products to the customer.

The stock records are updated to show that the items are now with the customer.

Customer Notifications

Automated reminders via email and/or portal alerts, sent N days before the scheduled rental return date.

End User Notifications

Email notifications to the end users N days before the scheduled rental return date, allowing them to prepare for pickup or follow-up.

The system sends timely reminders to both customers and internal users.

Customizable N value for notification lead time to match business needs.

Payment Gateway Integration

At least one secure payment gateway must be configured, enabling customers to make online payments directly from the rental quotation or order.

Popular gateways (e.g., PayPal, Stripe, Razorpay, etc.) can be integrated as per business requirements.

When you rent out products, the system automatically takes care of moving them through three main stages:

Reservation → Pickup → Return

Return (Collecting Back from the Customer)

When the rental period ends, the system generates a return document.

This guides your pickup team to collect the rented products from the customer.

Once returned, the stock records are updated so the items are available for the next rental.

4. Flexible Invoicing

When the Order is Confirmed, The user can create an initial invoice.

- Full upfront payment – Charge the total rental amount before the customer picks up the goods.
- Partial payment / deposit – Take a security deposit or first installment now, and charge the rest later.
- Late Return Fees: Automatically calculate late fees based on predefined rules.

5. Pricelist Management

Integrates its Pricelist system into the rental module to offer customized and flexible pricing.

Multiple Pricelists

- Create different pricelists for different customer segments, regions, or rental durations.
- E.g., Corporate clients, VIP customers, or seasonal discounts.

Time-Dependent Pricing

- Set variable pricing based on rental duration.
- Example: ₹10/hour, ₹60/day, ₹300/week – automatically applied based on rental period.

Discount Rules

- Apply percentage discounts, fixed discounts, or promotional offers to rental products.

Product Category or Customer-Specific Rules

- Define pricing rules per product category or specific customer groups.

Validity Periods

- Set date ranges for seasonal or promotional pricelists (e.g., “Holiday Rental Rates”).

6. Returns & Delays Handling

- Get alerts for late returns.
- Configure automatic late fees or penalties.

7. Reports and Dashboards

Tracking based on the given period:

- Most rented products
- Total rental revenue
- Top Customer

Downloadable Reports

All reports can be exported in multiple formats (PDF, XLSX, or CSV) for analysis, sharing, or record-keeping.

AI/GenAI Enhancement

The application should include an AI-powered rental component with the following additional functionalities:

1. **AI Rental Assistant:** Allow customers to describe their rental requirements in natural language, such as product type, rental duration, quantity, and budget. The system should recommend suitable available products.
2. **AI-Powered Product & Pricing Recommendations:** Based on product availability, rental duration, customer requirements, and existing pricing rules, the AI should suggest suitable products and an appropriate rental option.
3. **Conversational Rental Assistant:** Provide a conversational interface where users can ask questions about products, availability, pricing, rental policies, deposits, returns, and late fees, with responses based on the information available in the rental system.

Teams may use suitable LLMs, APIs, RAG, AI agents, or other GenAI approaches to implement these functionalities.